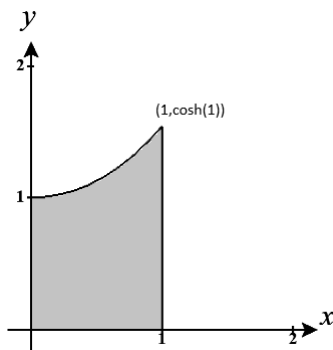


1.8 Hyperbolic Trigonometric Functions Worksheet

1. Consider the region R determined by the curves $y = \cosh(x)$, $x = 0$, $y = 0$, and $x = 1$.

a) Sketch a picture of the region R .

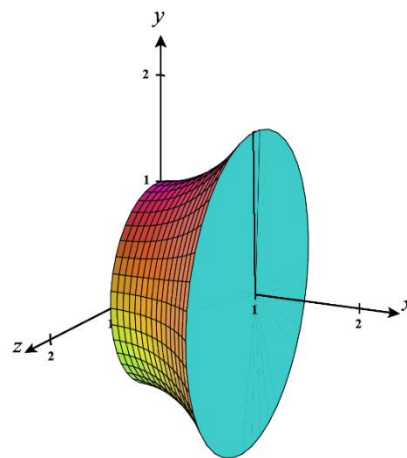


b) Set up, but do not evaluate, an integral which represents the area of the region R .

We note that $A = \int_0^1 [\cosh(x) - 0] dx$.

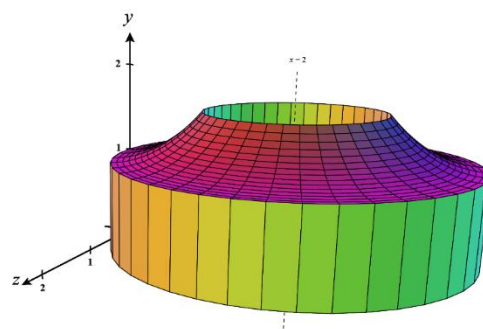
c) Suppose that R is revolved around the x -axis to create a solid. Setup, but do not evaluate, an integral which represents the volume of the solid.

$$V = \int_0^1 \pi [\cosh(x) - 0]^2 dx.$$



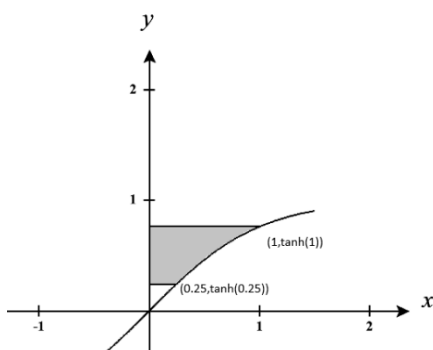
d) Suppose that R is revolved around the line $x = 2$ to create a solid. Setup, but do not evaluate, an integral which represents the volume of the solid.

$$V = \int_0^1 2\pi(2-x)(\cosh(x)-0) dx.$$



2. Consider the region R determined by the curves $y = \tanh(x)$, $y = \tanh(1)$, $y = \tanh(0.25)$, & $x = 0$.

a) Sketch a picture of the region R .

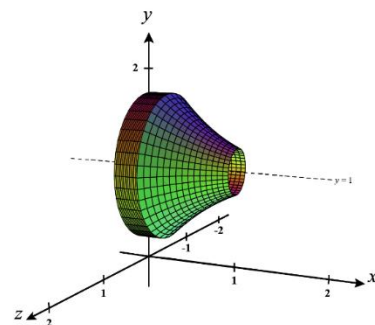


b) Set up, but do not evaluate, an integral(s) which represents the area of the region R .

$$A = \int_0^{0.25} [\tanh(1) - \tanh(0.25)] dx + \int_{0.25}^1 [\tanh(1) - \tanh(x)] dx$$

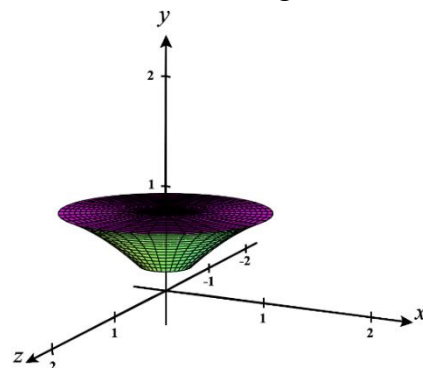
c) Suppose that R is revolved around the line $y = 1$ to create a solid. Setup, but do not evaluate, an integral which represents the volume of the solid.

$$V = \int_0^{0.25} \left[\pi(1 - \tanh(0.25))^2 - \pi(1 - \tanh(1))^2 \right] dx \\ + \int_{0.25}^1 \left[\pi(1 - \tanh(x))^2 - \pi(1 - \tanh(1))^2 \right] dx$$



d) Suppose that R is revolved around the y -axis to create a solid. Setup, but do not evaluate, an integral which represents the volume of the solid.

$$V = \int_0^{0.25} 2\pi(x-0) [\tanh(1) - \tanh(0.25)] dx \\ + \int_{0.25}^1 2\pi(x-0) [\tanh(1) - \tanh(x)] dx$$

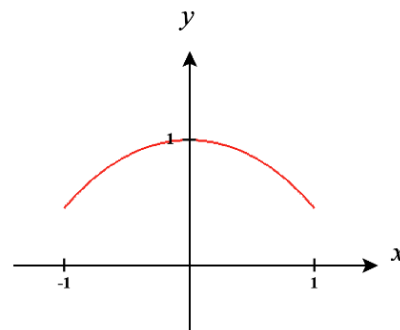


3. Consider the curve $y = 2 - \cosh(x)$ on $[-1, 1]$.

- a) Setup, but do not evaluate, an integral which represents the length of the curve.

We note that $y' = -\sinh(x)$ is continuous on $[-1, 1]$

$$L = \int_{-1}^1 \sqrt{1 + (-\sinh(x))^2} dx$$



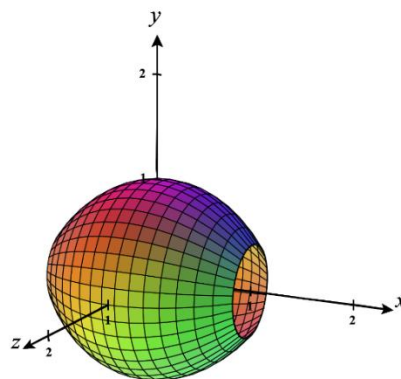
- b) Suppose the curve is rotated around the x -axis. Setup, but do not evaluate, an integral which represents the surface area of the resulting surface.

Note that $y = 2 - \cosh(x)$ is non-negative on $[-1, 1]$ and

$y' = -\sinh(x)$ is continuous on $[-1, 1]$.

Therefore, the surface area is given by

$$S = \int_{-1}^1 2\pi (2 - \cosh(x)) \sqrt{1 + (-\sinh(x))^2} dx.$$

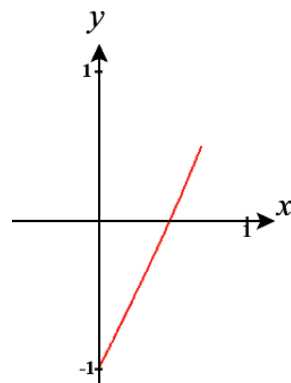


4. Consider the curve $y = 2 \sinh(x) - 1$ on $[0, \ln(2)]$.

- a) Setup, but do not evaluate, an integral which represents the length of the curve.

We note that $y' = 2 \cosh(x)$ is continuous on $[0, \ln(2)]$.

Therefore, the arc length is given by $L = \int_0^{\ln(2)} \sqrt{1 + (2 \cosh(x))^2} dx$.



- b) Suppose the curve is rotated around the x -axis. Why are we not able to use the surface area formula here to help find the surface area for the resulting surface?

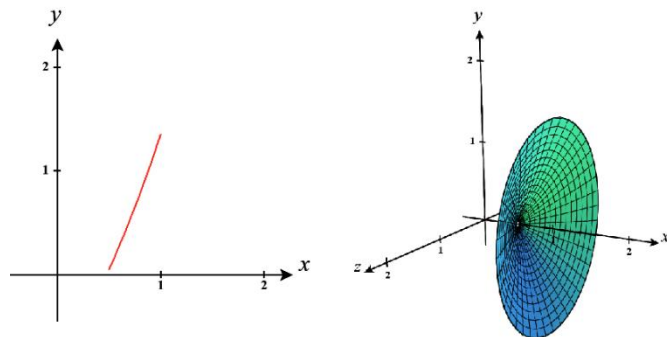
The issue here is that on the interval $[0, \ln(2)]$ the function $y = 2 \sinh(x) - 1$ is negative and therefore the conditions to use the surface area formula are not met.

- c) Suppose that we now consider the curve $y = 2\sinh(x) - 1$ on the interval $\left[\frac{1}{2}, 1\right]$. If this curve is rotated about the x -axis, then setup, but do not evaluate, an integral which represents the surface area of the resulting surface.

Note that $y = 2\sinh(x) - 1$ is non-negative on $\left[\frac{1}{2}, 1\right]$ and $y' = 2\cosh(x)$ is continuous on $\left[\frac{1}{2}, 1\right]$.

Therefore, the surface area is given by

$$S = \int_{1/2}^1 2\pi (2\sinh(x) - 1) \sqrt{1 + (2\cosh(x))^2} dx.$$



- d) Suppose that the same curve from part (c) is now revolved about the y -axis. Setup, but do not evaluate, an integral which represents the surface area of the resulting surface.

It isn't easy to solve our equation for x , so we will plan to set up this integral with respect to x . We already saw in part (c) that $y = 2\sinh(x) - 1$ is non-negative and $y' = 2\cosh(x)$ is continuous on $\left[\frac{1}{2}, 1\right]$.

Therefore, the surface area here is given by

$$S = \int_{1/2}^1 2\pi x \sqrt{1 + (2\cosh(x))^2} dx$$